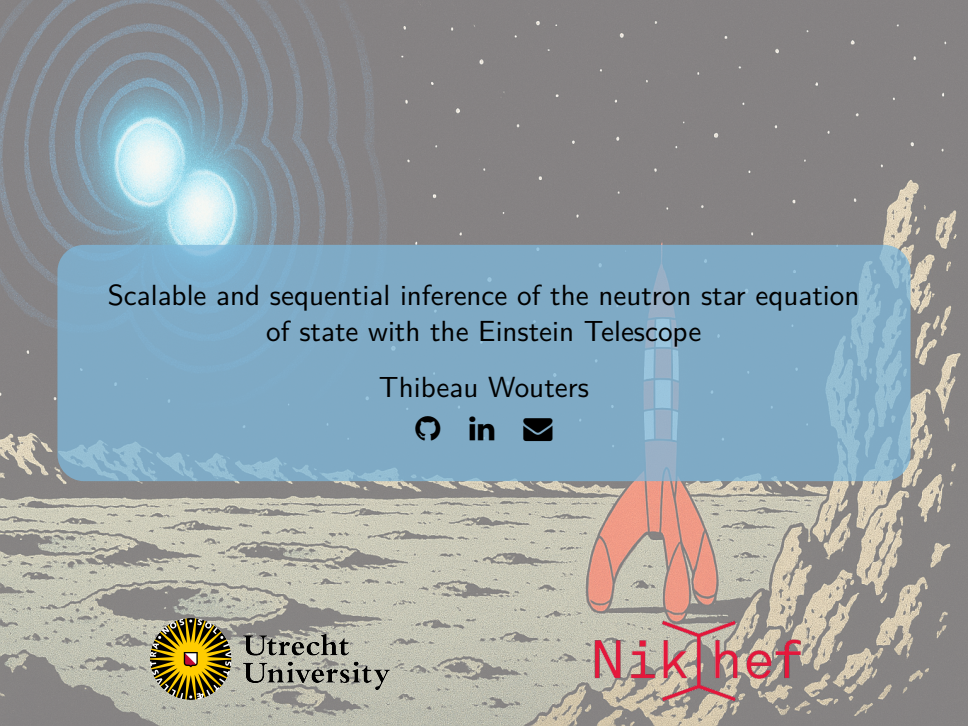




Scalable and sequential inference of the neutron star equation of state with the Einstein Telescope

ThibEAU Wouters



Utrecht
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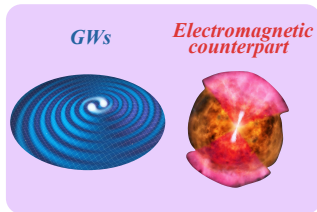
Neutron stars and fundamental physics

Neutron star mergers, especially when observed as multimessenger events (e.g., GW170817), probe

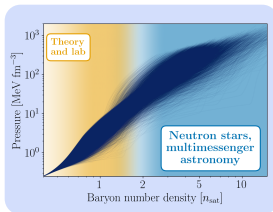
- Equation of state of ultradense matter [1]
- Expansion of the universe (H_0) [2]
- Tests of GR, populations, nucleosynthesis,...

data analysis

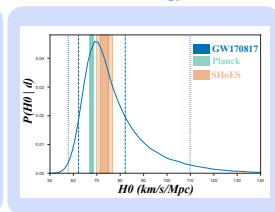
Multimessenger observations



Subatomic physics

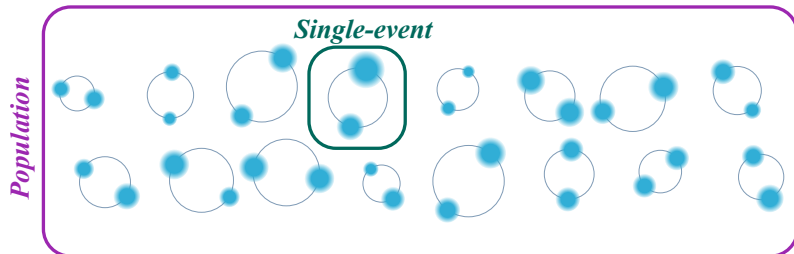


Cosmology

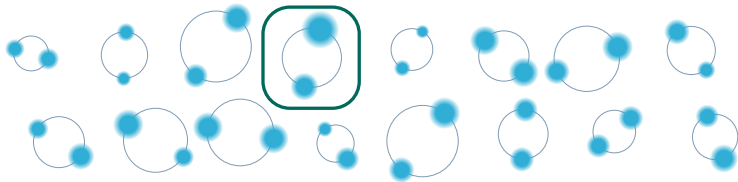


Challenges for Einstein Telescope

- Einstein Telescope
 - $\sim 30,000$ binary neutron star mergers/year
 - ~ 100 with detected electromagnetic counterpart
- **Challenge:** Data analysis (Bayesian inference) needs to scale up:
 - **Single-event:** analyze each event independently
 - **Population:** combine events into science insights
- **Solution:** GPU-acceleration and advanced algorithms



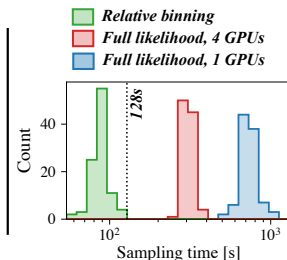
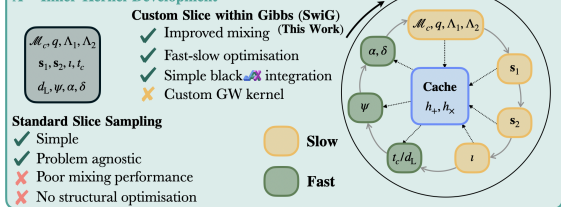
Single-event



Single event: slice-within-Gibbs sampler

- Standard sampling: move in full parameter space
- **Slice-within-Gibbs** [3]: move in ‘blocks’
 - **Slow**: new waveform generations, cached
 - **Fast**: projections, reuse cache
- Inference (for LVK signals!) in 15 mins with **no** approximation, or faster than signal duration (128 s) with relative binning

A – Inner Kernel Development



Towards ET?

- Development extended towards ET signals
 - RIPPLE 🌀: state-of-the-art waveforms (**talk by Robin**)
 - JIM 🌀: developing samplers (Thomas C. K. Ng)
- Strong BeNL presence (Harsh Narola, Peter T. H. Pang, Fabian Gittins, Isaac C. F. Wong, . . .)

New hardware-accelerated and accurate gravitational waveform approximants for fast parameter estimation

📅 Oct 9, 2026, 11:50 AM

🕒 20m

📍 Aula Magna



Robin Chan



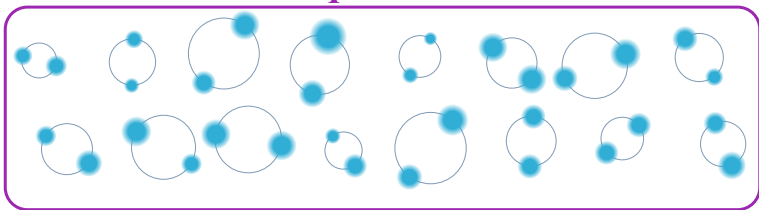
RIPPLE

Thomas C. K. Ng



JIM

Population

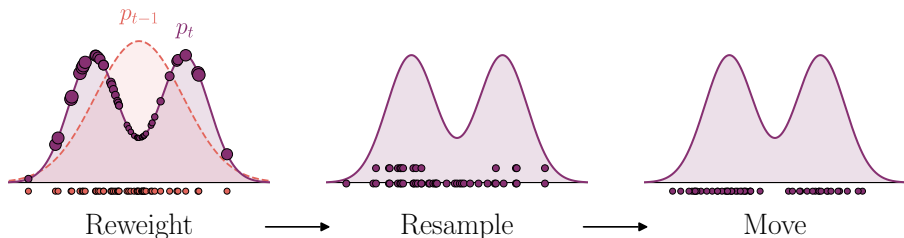


JESTER and sequential Monte Carlo (SMC)

- JESTER  [4]: used by LVK and ET
- From N events d_i to equation of state (EOS) posterior

$$\mathcal{P}(\theta_{\text{EOS}}|\{d_1, \dots, d_N\}) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta_{\text{EOS}}) \times \pi(\theta_{\text{EOS}})$$

- **Sequential Monte Carlo**: particles from prior $\rightarrow$ posterior through a sequence of intermediate densities p_t



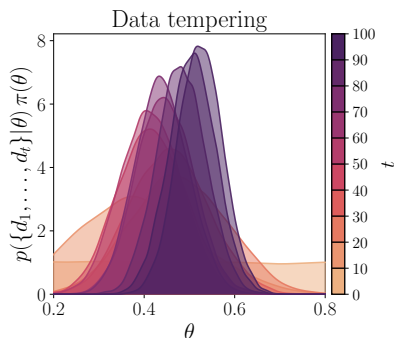
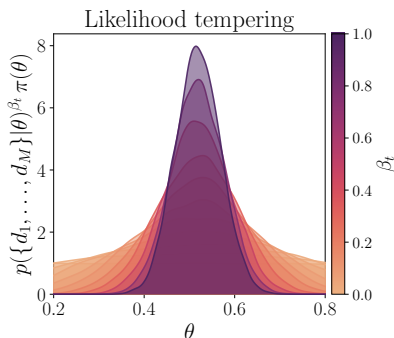
Likelihood vs. data tempering

① Likelihood tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_N\}|\theta)^{\beta_t} \pi(\theta) \quad \beta_t : 0 \rightarrow 1$$

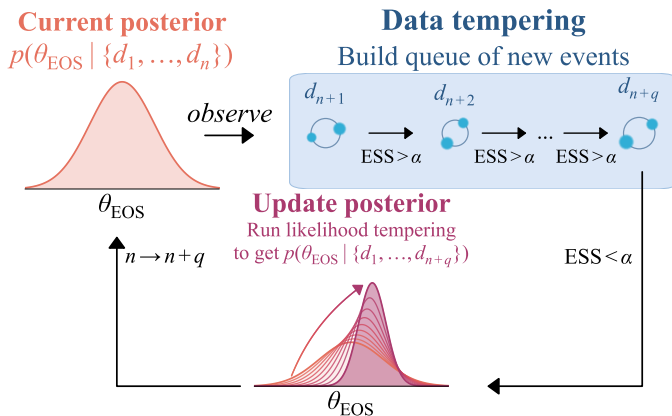
② Data tempering:

$$p_t(\theta) \propto \mathcal{L}(\{d_1, \dots, d_t\}|\theta) \pi(\theta) \quad t : 1 \rightarrow N$$



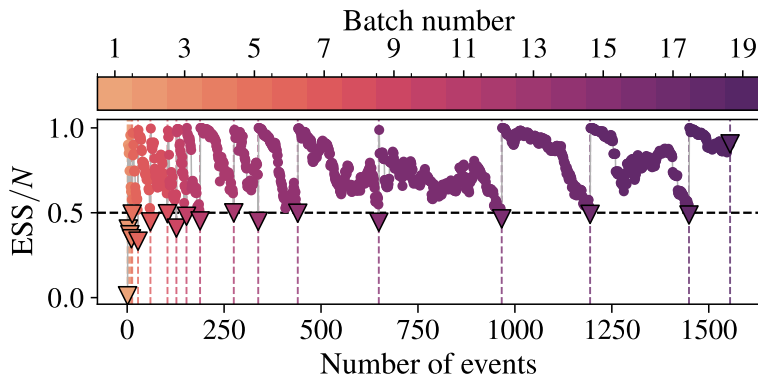
Adaptive hybrid algorithm

- ET catalogs will grow rapidly
- Need to sequentially update posteriors (data tempering)
- Use likelihood tempering for robustness



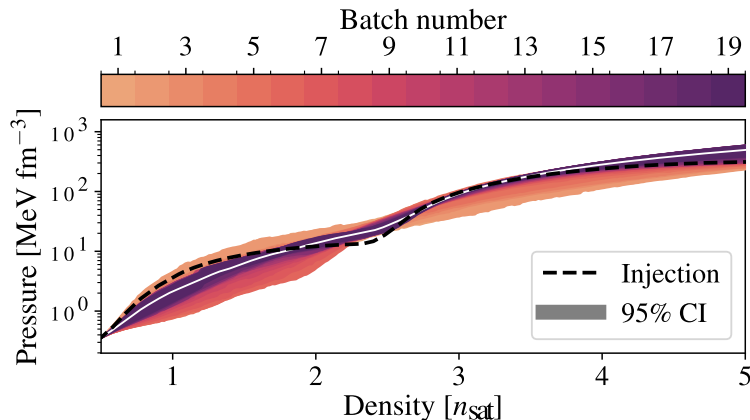
Proof of concept

- Toy catalogue: 1 month of binary neutron stars (GW) (~ 1500)
- Runtime ~ 2 hrs; expectation for 1 year: ~ 2 days



Proof of concept

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Multimessenger forecasts

- Multimessenger forecasts for population + equation of state + cosmology with JESTER [5]
- Runtime ~ 2 hrs; expectation for 1 year: ~ 2 days

Conclusions

- ET challenges our data analysis
- GPU $\times$ software innovations
 - **Single-event**: slice-within-Gibbs
 - **Population**: sequential Monte Carlo
- Results
 - **ripple, jim**: ET under development
 - **jester**
 - Scale to $\mathcal{O}(10^3)$ sources
 - Multimessenger forecasts

Thanks for listening!



References

- [1] B. P. Abbott et al. “GW170817: Measurements of neutron star radii and equation of state”. In: *Phys. Rev. Lett.* 121.16 (2018), p. 161101. DOI: [10.1103/PhysRevLett.121.161101](https://doi.org/10.1103/PhysRevLett.121.161101). arXiv: [1805.11581](https://arxiv.org/abs/1805.11581) [gr-qc].
- [2] B. P. Abbott et al. “A gravitational-wave standard siren measurement of the Hubble constant”. In: *Nature* 551.7678 (2017), pp. 85–88. DOI: [10.1038/nature24471](https://doi.org/10.1038/nature24471). arXiv: [1710.05835](https://arxiv.org/abs/1710.05835) [astro-ph.CO].
- [3] David Yallup et al. “Ab Initio Real-Time Gravitational-Wave Parameter Estimation”. In: (July 2026). arXiv: [2607.28265](https://arxiv.org/abs/2607.28265) [gr-qc].
- [4] Thibaud Wouters et al. “Leveraging differentiable programming in the inverse problem of neutron stars”. In: *Phys. Rev. D* 112.4 (2025), p. 043037. DOI: [10.1103/v2y8-kxvx](https://doi.org/10.1103/v2y8-kxvx). arXiv: [2504.15893](https://arxiv.org/abs/2504.15893) [astro-ph.HE].
- [5] Hauke Koehn et al. “Binary neutron stars in the next-generation era: Multi-messenger detection prospects and constraints on the equation of state, mass distribution, and cosmology”. In: (July 2026). arXiv: [2607.28438](https://arxiv.org/abs/2607.28438) [astro-ph.HE].

APPENDIX